Multivariate Calculus Lecture Notes
Section 13.1 – Introduction to Vector-Valued Functions

I. Idea of a vector-valued function
   a. A *vector-valued function* or *vector function* is a function that has vectors as outputs rather than scalars.
   b. Examples
      i. One input, 2D vector output: $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$
      ii. One input, 3D vector output: $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$
   c. We've already seen examples of these before; we called them *parametric equations*.

II. Sketching graphs of vector functions
   a. Sketching the graph of a vector function involves only graphing the ends of the vectors.
   b. The orientation of the graph is important.
   c. Examples
      i. $\mathbf{r}(t) = (1+t)\mathbf{i} + (2+5t)\mathbf{j} + (-1+6t)\mathbf{k}$, in 3D
         1. This is a line: $x = 1+t$, $y = 2+5t$, $z = -1+6t$.
         2. Take two points.
            - $\mathbf{r}(0) = \langle 1, 2, -1\rangle$
            - $\mathbf{r}(1) = \langle 2, 7, 5\rangle$
         3. Graphically:

      ii. $\mathbf{r}(t) = 2\cos t\mathbf{i} + \sin t\mathbf{j}$, in 2D
         1. Points
            - $\mathbf{r}(0) = \langle 2, 0\rangle$
            - $\mathbf{r}(\pi/2) = \langle 0, 1\rangle$
            - $\mathbf{r}(\pi) = \langle -2, 0\rangle$
            - $\mathbf{r}(3\pi/2) = \langle 0, -1\rangle$
         2. Graphically:

         3. This is an ellipse. (We've already seen this.)
      iii. $\mathbf{r}(t) = 2\cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$, in 3D
         1. This is the same graph as with the previous example, except that z is increasing over time.
         2. Graphically:

         3. This is a helix.
         4. Projections
            - Onto xy-plane: The same ellipse we saw before
            - Onto xz-plane: $x = 2\cos z$, which oscillates

- Onto yz-plane: $y = \sin z$, which also oscillates

III. Representing a graph by a vector function
   a. Parameterize $x = y^2 + 1$
      i. Let $y = t$, $x = t^2 + 1$.
      ii. We have $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + t\mathbf{j}$.
      iii. Graphically:

   b. Parameterizations
      i. What we just did is called a *parameterization* of a graph.
      ii. It's a way of representing the coordinates (x, y) of a graph as functions $x(t)$ and $y(t)$ in such a way that $(x(t), y(t))$ recreates the graph of the function as t ranges over all of its possible values.
   c. Sketch the intersection of the surfaces $z = x$ and $x = \sqrt{9 - y^2}$, and represent the curve as a vector function.
      i. Graphically:

      ii. Since z can be expressed in terms of x and x can be expressed in terms of y, it makes sense to let $y = t$.
      iii. This gives us $x = \sqrt{9 - t^2}$ and $z = \sqrt{9 - t^2}$.
      iv. The y-values run from -3 to 3, so this is the range of t values as well.
      v. Thus we get $\mathbf{r}(t) = \sqrt{9 - t^2}\mathbf{i} + t\mathbf{j} + \sqrt{9 - t^2}\mathbf{k}$, $-3 \leq t \leq 3$.
   d. (#25) Suppose we have the cylinders $x^2 + y^2 = 1$ and $x^2 + z^2 = 1$. (Use wooden demo?)
      i. Parameterize the intersection of these cylinders.
         1. First, let's think about what we're going to get.
            - Graphically:

            - The intersection will be two ellipses.
         2. We can parametrize the first cylinder as $\langle \cos t, \sin t, z \rangle$, as z can be anything.
         3. Solving for z in the second equation, we get $z = \pm\sqrt{1 - x^2} = \pm\sqrt{1 - \cos^2 t} = \pm \sin t$.
         4. So the parametrization is $\langle \cos t, \sin t, \pm \sin t \rangle$.
         5. The $\pm$ on the z coordinate captures the fact that there are two ellipses.
      ii. Describe the projection of this curve onto the three coordinate planes.
         1. Onto xy-plane: Circle of radius 1
         2. Onto xz-plane: Circle of radius 1
         3. Onto yz-plane: Two intersecting line segments: $z = \pm y$, for $-1 \leq y \leq 1$

IV. Domain of a vector function

a. The domain of a vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is the intersection of the domains of $f(t)$, $g(t)$, and $h(t)$.
b. Example: $\mathbf{r}(t) = t^3\mathbf{i} + \ln(3-t)\mathbf{j} + \sqrt{t}\mathbf{k}$.
   i. Domain of t^3: $\mathbb{R}^n$
   ii. Domain of $\ln(3-t)$: $3 - t > 0 \implies t < 3$
   iii. Domain of $\sqrt{t}$: $t \geq 0$
   iv. Domain of $\mathbf{r}(t)$: $[0, 3)$

Homework: Section 13.1 #2, 12, 18, 21, 27, 32, 33, 36, 38, 41.